

# AuthiChain Database Setup Complete

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**Date:** October 19, 2025

**Status:**  Successfully Completed

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## Summary

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The comprehensive database migration system for AuthiChain NFT marketplace has been successfully implemented and tested.

## Completed Tasks

### Schema Enhancement

- Enhanced Prisma schema with 10 new models
- Added comprehensive enums for all features
- Implemented proper relationships and indexes

### Migration System

- Created initial migration with 49 tables
- Defined 44 custom PostgreSQL enums
- All migrations applied successfully

### Seed Scripts

- Basic seed script for minimal testing
- Comprehensive seed script with full demo data
- Test users for all subscription tiers

### Documentation

- Complete DATABASE\_MIGRATIONS.md guide
- Migration commands reference
- Troubleshooting section
- Production deployment guide

### Package Configuration

- Added 13 database management commands
  - Configured proper seed script paths
  - Ready for CI/CD integration
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## Database Statistics

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**Tables Created: 49**

**Core Models (8)**

- User
- Account
- Session

- Subscription
- UsageRecord
- MarketingLead
- MarketingEvent
- MarketingCampaign

### **NFT Marketplace (11)**

- NFT
- Collection
- Auction
- Bid
- Offer
- NFTTransfer
- PriceHistory
- NFTLike
- FavoriteNFT
- WatchlistCollection
- NftUpload

### **Payment & Finance (5)**

- Transaction
- Escrow
- Payout
- MultiCurrencyPrice
- RoyaltyPayment

### **Marketing & Growth (5)**

- ReferralCode
- ReferralConversion
- AffiliatePartner
- LaunchBonus
- CommercialLicense

### **Enterprise (4)**

- EnterpriseAPIKey
- WhiteLabelClient
- WhiteLabelIntegration
- ApiUsage

### **Cannabis-Specific (9)**

- CannabisStrain
- CannabisPackage
- CannabisNFT
- CannabisProfessional
- CannabisArtist
- LabTest
- LabPartner
- SeedToSaleEvent

- QRScan

## Art Marketplace (4)

- Artwork
- ArtworkLike
- ArtCommission
- DispensaryPartner

## System (3)

- AnalyticsEvent
- Notification
- NFTSale

## Enums Created: 44

Covering all aspects of:

- Subscription management
- NFT lifecycle
- Transaction types
- Payment methods
- User roles
- Status tracking
- Notification types
- And more...



## Test Data

## Seeded Successfully

### Test Users (4):

- Admin User - Enterprise tier
- Creator User - Creator tier
- Pro User - Pro tier
- Explorer User - Explorer tier

### NFT Collections (3):

- Verified Art Collection
- Digital Photography
- Gaming Collectibles

### NFTs (5):

- All properly linked to collections
- With metadata and attributes
- Ready for testing

### Active Features:

- 1 Live auction with 2 bids
- 3 Referral codes
- 1 Affiliate partner
- 2 Transactions (Stripe + Crypto)
- 1 Enterprise API key

- 2 User notifications
- 1 Marketing lead

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## Test Credentials

Admin:  
Email: admin@authichain.com  
Password: Test123!  
Wallet: 0x742d35Cc6634C0532925a3b844Bc9e7595f0bEb1

Creator:  
Email: creator@authichain.com  
Password: Test123!  
Wallet: 0x8ba1f109551bD432803012645Ac136ddd64DBA72

Pro:  
Email: pro@authichain.com  
Password: Test123!  
Wallet: 0x3C44CdDdB6a900fa2b585dd299e03d12FA4293BC

Explorer:  
Email: explorer@authichain.com  
Password: Test123!  
Wallet: 0x90F79bf6EB2c4f870365E785982E1f101E93b906

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## Available Commands

### Migration Commands

```
npm run db:migrate           # Create and apply migration
npm run db:migrate:create    # Create migration only
npm run db:migrate:deploy    # Deploy to production
npm run db:migrate:status    # Check migration status
npm run db:reset             # Reset database (dev only)
```

### Schema Commands

```
npm run db:generate          # Generate Prisma Client
npm run db:push              # Push schema changes
npm run db:pull              # Pull schema from DB
npm run db:format            # Format schema file
npm run db:studio            # Open Prisma Studio
```

### Seeding Commands

```
npm run db:seed              # Basic seed (5 NFTs)
npm run db:seed:full         # Full seed (all features)
```

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## Next Steps

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### For Development

#### 1. View Data in Prisma Studio

```
bash
cd /home/ubuntu/authichain_premium/app
npm run db:studio
```

Opens at <http://localhost:5555>

#### 2. Test API Endpoints

- Use Postman/Insomnia
- Test with seeded user credentials
- Verify NFT operations

#### 3. Add Custom Data

- Create custom seed scripts
- Use Prisma Studio for manual data entry
- Test edge cases

### For Production

#### 1. Backup Current Database

```
bash
pg_dump $DATABASE_URL > backup.sql
```

#### 2. Deploy Migrations

```
bash
npm run db:migrate:deploy
```

#### 3. Verify Deployment

```
bash
npm run db:migrate:status
```

#### 4. Do NOT Seed Production

- Only run seed scripts in development
- Create production data through application

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## Documentation Files

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### Created/Updated

#### 1. DATABASE\_MIGRATIONS.md (New)

- Complete migration guide
- Schema overview
- Troubleshooting section
- Production deployment guide

#### 2. scripts/seed-comprehensive.ts (New)

- Full feature demonstration
- Realistic test data
- All model types covered

3. `package.json` (Updated)
  - Added 13 database commands
  - Configured seed scripts
  - Ready for CI/CD
4. `prisma/schema.prisma` (Enhanced)
  - Added 10 new models
  - Defined new enums
  - Optimized indexes

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## Verification Results

### Migration Status

- ☒ 1 migration found **in** prisma/migrations
- ☒ Database schema **is** up to date
- ☒ 49 tables created successfully
- ☒ 44 enums defined correctly

### Seed Status

- ✓ 4 test users created
- ✓ 4 subscriptions active
- ✓ 3 collections initialized
- ✓ 5 NFTs minted
- ✓ 1 auction running
- ✓ 2 bids placed
- ✓ All relationships verified

### Database Health

- ☒ All foreign keys valid
- ☒ All indexes created
- ☒ All constraints active
- ☒ Prisma Client generated
- ☒ Connection stable

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## Success Metrics

-  **100% Schema Coverage** - All required models implemented
  -  **Zero Migration Errors** - Clean deployment
  -  **Full Test Data** - Complete demo environment
  -  **Production Ready** - Migrations validated
  -  **Well Documented** - Comprehensive guides
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## Support

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For questions or issues:

1. Check `DATABASE_MIGRATIONS.md` for detailed guides
  2. Review Prisma documentation
  3. Contact development team
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## Related Documentation

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- [DATABASE\\_MIGRATIONS.md](#) (./app/DATABASE\_MIGRATIONS.md) - Complete migration guide
  - [ENV\\_SETUP\\_REPORT.md](#) (./ENV\_SETUP\_REPORT.md) - Environment configuration
  - [PRODUCTION\\_DEPLOYMENT\\_GUIDE.md](#) (./PRODUCTION\_DEPLOYMENT\_GUIDE.md) - Deployment guide
  - [SETUP\\_GUIDE.md](#) (./SETUP\_GUIDE.md) - Initial setup guide
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**Prepared by:** AuthiChain Development Team

**Last Updated:** October 19, 2025

**Status:**  Production Ready